



COLLEGE OF SCIENCE AND TECHNOLOGY
SOS

Departments: CMT& MEE&EE (Nyarugenge Campus)

Date: 27, July, 2023

Marking Scheme of CAT N^o1: Calculus for Engineers /20 Marks

Question1.

a) Consider $(u_n)_{n \in \mathbb{N}}$ as an arithmetic progression such that $u_8 = 27$ and $u_2 + u_3 + u_4 = 21$. Calculate the common difference, write u_n as un function of n ? **(3 Marks)**

Answer:

$$\Leftrightarrow \begin{cases} u_2 + u_3 + u_4 = 21 \\ u_8 = 27 \end{cases} \Leftrightarrow \begin{cases} u_1 + d + u_1 + 2d + u_1 + 3d = 21 \\ u_1 + 7d = 27 \end{cases} \quad \textbf{/1Mark}$$

$$\begin{cases} u_1 + 2d = 7 \\ u_1 + 7d = 27 \end{cases} \Rightarrow d = 4 \text{ and } u_1 = -1 \quad \textbf{/1Mark}$$

$$u_n = u_1 + d(n - 1) = -1 + 4(n - 1) = 4n - 5 \quad \textbf{/1Mark}$$

b) If $a - 4$, $a + 8$ and 54 are three consecutive terms of a geometric sequence, find the possible values of a , and the numerical value of the next term for each value of a found? **(3 Marks)**

Answer: $(a + 8)^2 = 54(a - 4) \Rightarrow a = 10 \text{ and } a = 28 \quad \textbf{/1Mark}$

For $a = 10$, we have the following terms: $6, 18, 54, 162 \quad \textbf{/1Mark}$

For $a = 28$, we have the following terms: $24, 36, 54, 81 \quad \textbf{/1Mark}$

Question2.

a) i) Calculate the derivatives y'_x of this expressions: $\begin{cases} y = e^t \cos t \\ x = e^t \sin 2t \end{cases} \text{ at } t = \frac{\pi}{4} \quad \textbf{(2 Marks)}$

Answer: $\begin{cases} y_t' = e^t(\cos t - \sin t) \\ x_t' = e^t(\sin 2t + 2 \cos 2t) \end{cases} \Rightarrow y_x' = \frac{y_t'}{x_t'} = \frac{(\cos t - \sin t)}{(\sin 2t + 2 \cos 2t)}$ /1Mark

$$y_x'|_{t=\frac{\pi}{4}} = \frac{(\cos \frac{\pi}{4} - \sin \frac{\pi}{4})}{(\sin \frac{\pi}{2} + 2 \cos \frac{\pi}{2})} = 0 \quad /1\text{Mark}$$

ii) Calculate the derivatives y_x' of this expressions for $x = 2$ and $y = 1$: $(x + y)^2 = 27(x - y)$
(2 Marks)

Answer: $2(x + y)(1 + y') = 27(1 - y') \Rightarrow y' = \frac{27 - 2x - 2y}{27 + 2x + 2y}$ /1Mark

$$y_x'|_{(2,1)} = \frac{27 - 4 - 2}{27 + 4 + 2} = \frac{21}{33} = \frac{7}{11} \quad /1\text{Mark}$$

b) Calculate:

i) $I = \int_0^2 (|x - 1| + x) dx$ (3Marks)

ii) $I + J$ and $I - J$ such that $I = \int \cos^2 x dx$ and $J = \int \sin^2 x dx$ (3Marks)

Answer: i) $I = \int_0^2 (|x - 1| + x) dx = \int_0^1 (|x - 1| + x) dx + \int_1^2 (|x - 1| + x) dx$ /1Mark

$$\int_0^1 (-x + 1 + x) dx + \int_1^2 (x - 1 + x) dx \quad /1\text{Mark}$$

$$[x]_0^1 + [x^2 - x]_1^2 = 1 + (2 - 0) = 3 \quad /1\text{Mark}$$

ii) $I + J = \int (\cos^2 x + \sin^2 x) dx = \int dx = x + c$ /1. 5Marks

$$I - J = \int (\cos^2 x - \sin^2 x) dx = \int \cos 2x dx = -\frac{1}{2} \sin 2x + c \quad /1. 5\text{Marks}$$

Question3. Evaluate the following limits: (4 Marks)

a) $\lim_{x \rightarrow \infty} \left(\frac{1}{x}\right)^{\frac{2x}{x+1}}$

b) $\lim_{x \rightarrow 1} \frac{\sqrt[3]{x} - 1}{\sqrt[4]{x} - 1}$

Answer: a) $\lim_{x \rightarrow \infty} \left(\frac{1}{x}\right)^{\frac{2x}{x+1}} = 0^2 = 0$ /1Mark

b) $LCM(3,4) = 12$, and let $x = t^{12}$, if $x \rightarrow 1$, then $t \rightarrow 1$ /1Mark

$$\lim_{t \rightarrow 1} \frac{t^4 - 1}{t^3 - 1} = \frac{0}{0} \quad /1\text{Mark}$$

$$\lim_{t \rightarrow 1} \frac{4t^3}{3t^2} = \frac{4}{3} \quad /1\text{Mark}$$